

## 13. Coding & Solution

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| <b>Date</b>          | 30 June 2026                                        |
| <b>Team ID</b>       | SWTID-2026-4570                                     |
| <b>Project Name</b>  | EduGenie – Google Gemini Powered Learning Assistant |
| <b>Maximum Marks</b> | 5 Marks                                             |

### Solution Summary

|                                      |                                                                                                                                                                                                                                          |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Repository Link / URL</b>         | <a href="https://github.com/Saajith-07/EduGenie-Google-Gemini-Powered-Learning-Assistant">https://github.com/Saajith-07/EduGenie-Google-Gemini-Powered-Learning-Assistant</a>                                                            |
| <b>Programming Language(s)</b>       | Python 3.10, JavaScript, HTML, CSS                                                                                                                                                                                                       |
| <b>Framework(s) Used</b>             | FastAPI, Uvicorn, Jinja2                                                                                                                                                                                                                 |
| <b>Key Features Implemented</b>      | AI Q&A chat; leveled concept explanation (local T5 + Gemini fallback); MCQ quiz generation; text/notes summarization; personalized learning roadmap; JSON-file query/response logging; demo/mock mode; Dockerized deployment             |
| <b>Pending / Incomplete Features</b> | User authentication and role-based access; migration from JSON files to a relational database; usage analytics dashboard; consolidation of duplicate router/direct API endpoints                                                         |
| <b>Setup / Run Instructions</b>      | <p>1) pip install -r requirements.txt 2) Copy .env.example to .env and set GEMINI_API_KEY 3) uvicorn main:app --reload 4) Open http://127.0.0.1:8000</p> <p>Docker: docker build -t edugenie . then docker run -p 8000:8000 edugenie</p> |

### Code Quality Checklist

| S.No | Criteria                                               | Status                                                                      |
|------|--------------------------------------------------------|-----------------------------------------------------------------------------|
| 1    | Code is modular and organized into functions / classes | Yes                                                                         |
| 2    | Meaningful variable and function names are used        | Yes                                                                         |
| 3    | Code includes comments / documentation where necessary | Yes                                                                         |
| 4    | Error handling is implemented for critical operations  | Yes                                                                         |
| 5    | The application runs without critical errors           | Yes (verified via tests/test_api.py, all routes return 200/422 as expected) |
| 6    | Code is committed to a version control repository      | Yes (GitHub)                                                                |

### **Additional Notes / Comments**

The application was verified locally in demo/mock mode (no GEMINI\_API\_KEY configured) using FastAPI's TestClient — all page routes, the five router-based endpoints and the five direct endpoints returned successful responses. CORS is currently open to all origins for local testing and should be restricted before any public deployment.